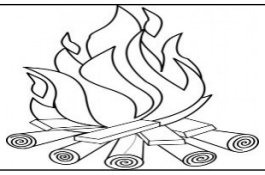
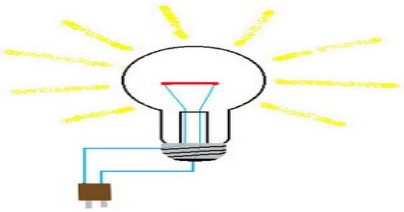
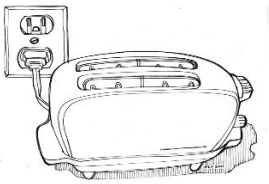
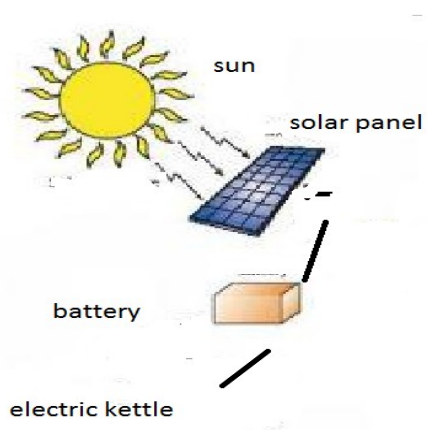
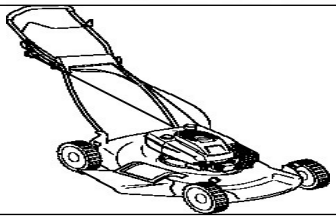
Energy Transformation

Energy transformation= Where _____ changes from one type of _____ to another.

We can show the energy transformations in an energy flow diagram

Types of energy that you can use to make the flow diagrams are, chemical, electric, light, heat, sound, solar, kinetic.

Situation	Picture	Energy flow diagram
		Chemical → Heat Light
Electric light		Electric
toaster		
Mobile phone		
Solar panel Charging a battery And Being used to Boil a kettle		
Lawn mover		

Motor car

Petrol stores energy chemical potential

Petrol burns heat

Engine moves kinetic

Heat

sound

Car moves kinetic

Did all the energy from the store of energy in the petrol get used to move the car?

What else did the stored energy get used for?

Were these other forms of energy made very useful?

We call the energy that is **used** from the petrol the INPUT ENERGY.

We call the **useful** energy of the car moving the OUTPUT ENERGY.

The non-useful heat and sound energy are a waste of energy.

Energy efficiency is a measure of how much of in the input energy gets changed into USEFUL OUTPUT ENERGY.

$$\text{Energy efficiency} = \frac{\text{useful energy output}}{\text{Energy input}} \times 100$$

We can calculate the energy efficiency of different things.

For example a petrol car uses 10,000kj of energy to get 2,500kj of forward kinetic movement.

What is its efficiency?

$$\text{Energy efficiency} = \frac{\text{useful energy output}}{\text{Energy input}} \times 100$$

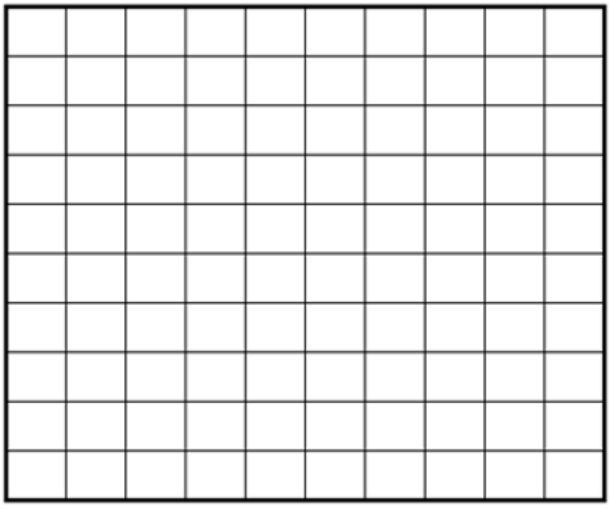
$$\text{Energy efficiency} = \frac{2,500}{10,000} \times 100$$

$$= 25\%$$

Complete this table

Situation	Energy input (kj)	Useful energy output (kj)	calculation	Efficiency (%)
Boy riding a bike	1000	950	_____ x 100	
Petrol car	20,000	5,000	_____ x 100	
Diesel car	20,000	7,000	_____ x 100	
Steam train	100,000	40,000	_____ x 100	

Comparing the Energy Efficiency of Different types of Transport



Different types of Transport